FUEL EFFICIENCY PREDICTION USING MACHINE LEARNING

PROJECT SYNOPSIS
of Major Project

BACHELOR OF TECHNOLOGY

Computer Science & Engineering

SUBMITTED BY   GUIDED BY:

Name: Sakshi (2300466)
Name: Sandeep Kumar Yadav (2200468)
Name: Sahil Dhiman (2300465)
Name: Sahil (2300464)

Baba Banda Singh Bahadur Engineering College Fatehgarh Sahib

February 2026

INDEX

CHAPTER 1. INTRODUCTION

1. Project Overview

Fuel efficiency prediction has become an essential aspect of the automotive industry due to rising fuel costs, increasing vehicle usage, and growing environmental concerns. Fuel efficiency represents the distance traveled by a vehicle per unit of fuel consumed and is commonly measured in miles per gallon (MPG) or kilometers per liter (km/l). Accurate prediction of fuel efficiency helps vehicle buyers make informed decisions, assists manufacturers in optimizing vehicle design, and supports efforts to reduce fuel consumption and carbon emissions. The **Fuel Efficiency Prediction using Machine Learning** project aims to develop an intelligent predictive system that estimates vehicle fuel efficiency based on various vehicle characteristics such as engine displacement, number of cylinders, horsepower, weight, acceleration, model year, and origin of manufacture. By leveraging machine learning techniques, the system learns complex and non-linear relationships between these parameters and fuel efficiency. The proposed system provides accurate predictions along with visual analysis and real-time user interaction through a web-based interface, making it suitable for both academic and practical applications.

1.1 Technology Used:

Frontend: Streamlit for developing an interactive, user-friendly web interface for real-time prediction and visualization.
Backend: Python for implementing machine learning algorithms and data processing workflows
Machine Learning Libraries:
Scikit-learn for regression models and evaluation
Pandas and NumPy for data preprocessing and manipulation
Visualization Tools: Matplotlib and Seaborn for graphical analysis and result interpretation
Model Storage and Deployment: Joblib for saving and loading trained machine learning models

1.2 Field of the Project:

This project lies at the intersection of Machine Learning, Data Science, and Automotive Engineering. It is a specialized predictive analytics solution focused on vehicle performance evaluation, fuel optimization, and sustainable transportation systems.

2. Key Technical Terms and Special Features:

- **Key Technical Terms**

1. **Machine Learning (ML):**
   branch of artificial intelligence that enables systems to learn patterns from data and make predictions without explicit programming.
2. **Supervised Learning:**

Machine learning approach where models are trained using labelled datasets containing input features and corresponding output values.

3. **Regression Analysis:**
   A technique used to predict continuous numerical values such as fuel efficiency.
4. **Feature Engineering:**
   The process of selecting, transforming, and optimizing input variables to improve model accuracy.
5. **Model Evaluation Metrics:**
   Statistical measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score used to assess prediction performance.

- **Special Features**
  1. **Multiple Model Comparison:**
     The system implements and compares multiple machine learning models such as Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor to select the most accurate prediction model.
  2. **Advanced Feature Engineering:**
     Correlation analysis, normalization, encoding of categorical data, and feature importance analysis are performed to enhance prediction reliability.
  3. **Real-Time Prediction Interface:**
     A web-based front-end allows users to input vehicle specifications and obtain instant fuel efficiency predictions.
  4. **Indian Metric Conversion:**
     The predicted fuel efficiency is displayed in both MPG and km/l for better regional understanding and usability.
  5. **Visual Graphs and Analysis:**
     Graphical representations such as actual vs. predicted values and feature importance charts provide interpretability and insight into the model's behavior.

3. Purpose and Vision:

Purpose:
The purpose of this project is to develop a machine learning–based system that accurately predicts vehicle fuel efficiency using key vehicle parameters. The system aims to support informed decision-making by providing reliable and real-time fuel efficiency predictions in a user-friendly manner.
Vision:
The vision of this project is to promote sustainable and fuel-efficient transportation by leveraging machine learning and data analytics, thereby reducing fuel consumption, environmental impact, and supporting smarter automotive choices.

CHAPTER 2: LITERATURE SURVEY

1. Introduction

A literature survey helps in understanding existing research, methodologies, and
technologies related to the project domain. In the area of vehicle fuel efficiency prediction, several researchers have applied statistical and machine learning techniques to estimate fuel efficiency based on vehicle characteristics. Recent studies indicate that machine learning models provide better accuracy than traditional approaches by capturing complex relationships among vehicle parameters.

2. Existing Similar Projects and Initiatives

This section reviews key research works and projects related to fuel efficiency prediction using machine learning techniques.

2.1 Machine Learning-Based Fuel Efficiency Prediction (Scientific Reports, Nature, 2025)

Overview: A study published in *Scientific Reports (Nature)* proposed a machine learning framework for predicting vehicle fuel efficiency using multiple vehicle attributes. The study compared several regression models to identify the most
accurate prediction approach.
Key Features:
–Use of ensemble models such as Random Forest and Gradient Boosting
–Evaluation using MAE, RMSE, and R^2 Score

2.2 Regression-Based Fuel Efficiency Prediction Models

Overview: Many researchers have used regression techniques such as Linear Regression and Support Vector Regression with datasets like Auto MPG to predict fuel efficiency. These models serve as baseline approaches for understanding the relationship between vehicle parameters and fuel efficiency.
Key Features:
Simple and interpretable models
Limited accuracy due to linear assumptions

2.3 Feature Engineering and Model Evaluation Studies

Overview: Recent research emphasizes the role of data preprocessing and feature engineering in improving prediction accuracy. Techniques such as normalization, correlation analysis, and feature selection significantly enhance model performance.

Key Features:
Improved accuracy through feature optimization
Better understanding of influential vehicle parameters

2.4 GPS-Based Machine Learning Approaches (MDPI, 2025)

Overview: A study published in *Sustainability (MDPI)* used GPS-based real-world driving data to predict fuel-related metrics using machine learning models.
Key Features:
Use of ensemble and decision tree-based models
Application of real driving data for prediction

2.5 Machine Learning Models for Fuel Consumption and Efficiency Prediction

Overview: Several studies have applied machine learning algorithms such as Random Forest, Support Vector Regression, and Gradient Boosting to predict fuel consumption and fuel efficiency.
Key Features:
Improved accuracy over traditional regression methods
Ability to capture non-linear relationships among vehicle parameters

3. Research Gap

Although existing studies achieve good prediction accuracy, many lack real-time deployment and user interaction. Additionally, limited focus has been placed on visual analysis and regional metric conversion. The proposed project addresses these gaps by integrating multiple model comparison, visual analysis, real-time prediction, and km/l conversion for practical usability.

CHAPTER 3: METHODOLOGY

1. Requirement Analysis and Research

The project requirements were analysed to define the scope and objectives of predicting vehicle fuel efficiency using machine learning. Research was conducted on existing fuel efficiency prediction studies and publicly available datasets to identify suitable machine learning techniques, evaluation metrics, and deployment methods.

2. System Design and Architecture

The system architecture was designed to follow a modular and layered approach. It consists of data preprocessing, machine learning model training, evaluation, and deployment layers. The overall architecture ensures smooth data flow from raw dataset input to final fuel efficiency prediction output. User interface mockups were designed to visualize how users would interact with the system for entering vehicle parameters and viewing results. The selected technology stack is as follows:

- Frontend: Streamlit for developing an interactive and user-friendly web interface.
- Backend Core Logic: Python for handling data processing, model training, and prediction logic.
- Data Storage: CSV-based dataset and serialized trained models using Joblib.

3. Data Collection and Preprocessing

Vehicle data was collected from standard and reliable datasets such as the Auto MPG dataset. The collected data included attributes like engine displacement, horsepower, weight, number of cylinders, acceleration, model year, and origin. Data preprocessing was performed to improve data quality and model performance.

- Missing values were handled appropriately.
- Categorical variables were encoded using suitable encoding techniques.
- Numerical features were normalized and scaled.
- Outliers were analyzed and treated where necessary.

4. Machine Learning Model Development

- Machine learning techniques were applied to develop predictive models for fuel efficiency estimation.
- Multiple regression models such as Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor were implemented.
- The dataset was split into training and testing sets to evaluate model generalization.

- Feature engineering techniques such as correlation analysis and feature selection were applied to enhance model accuracy.
- The use of multiple models allowed performance comparison and selection of the most accurate prediction model.

5. Model Evaluation and Analysis

To assess the effectiveness of the trained models, various evaluation metrics were used:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Graphical analysis such as actual versus predicted plots and feature importance graphs were generated to interpret model behaviour and validate prediction reliability.

6. Front-End Interface and Prediction System

A web-based front-end interface was developed using Streamlit to enable real-time interaction with the trained machine learning model.

- Users can input vehicle specifications through the interface.
- The trained model processes the input data and predicts fuel efficiency instantly.
- Predicted results are displayed in both MPG and Indian-relevant units (km/l).
- Visual graphs are presented to enhance result interpretation.

This interface makes the system practical and easy to use for non-technical users.

7. Testing and Validation

To ensure reliability and correctness, multiple testing techniques were applied:

- Unit testing for data preprocessing and prediction functions.
- Integration testing to ensure smooth interaction between the model and user interface.
- System testing to validate overall system performance.
- Validation using unseen test data to ensure prediction accuracy and robustness.

8. Deployment and Documentation

After successful testing, the trained model was deployed using the Streamlit platform for demonstration and real-time usage.

- The model was saved using Joblib for efficient reuse.

- The application was tested for different input scenarios.
- Proper documentation and a user guide were prepared to explain system usage and functionality.

9. Final Report and Presentation

The final phase involved documenting the complete project lifecycle, including system design, methodology, implementation, and results. A project demonstration was prepared to showcase real-time fuel efficiency prediction, visual analysis, and model performance comparison, highlighting the practical applicability of the proposed system.

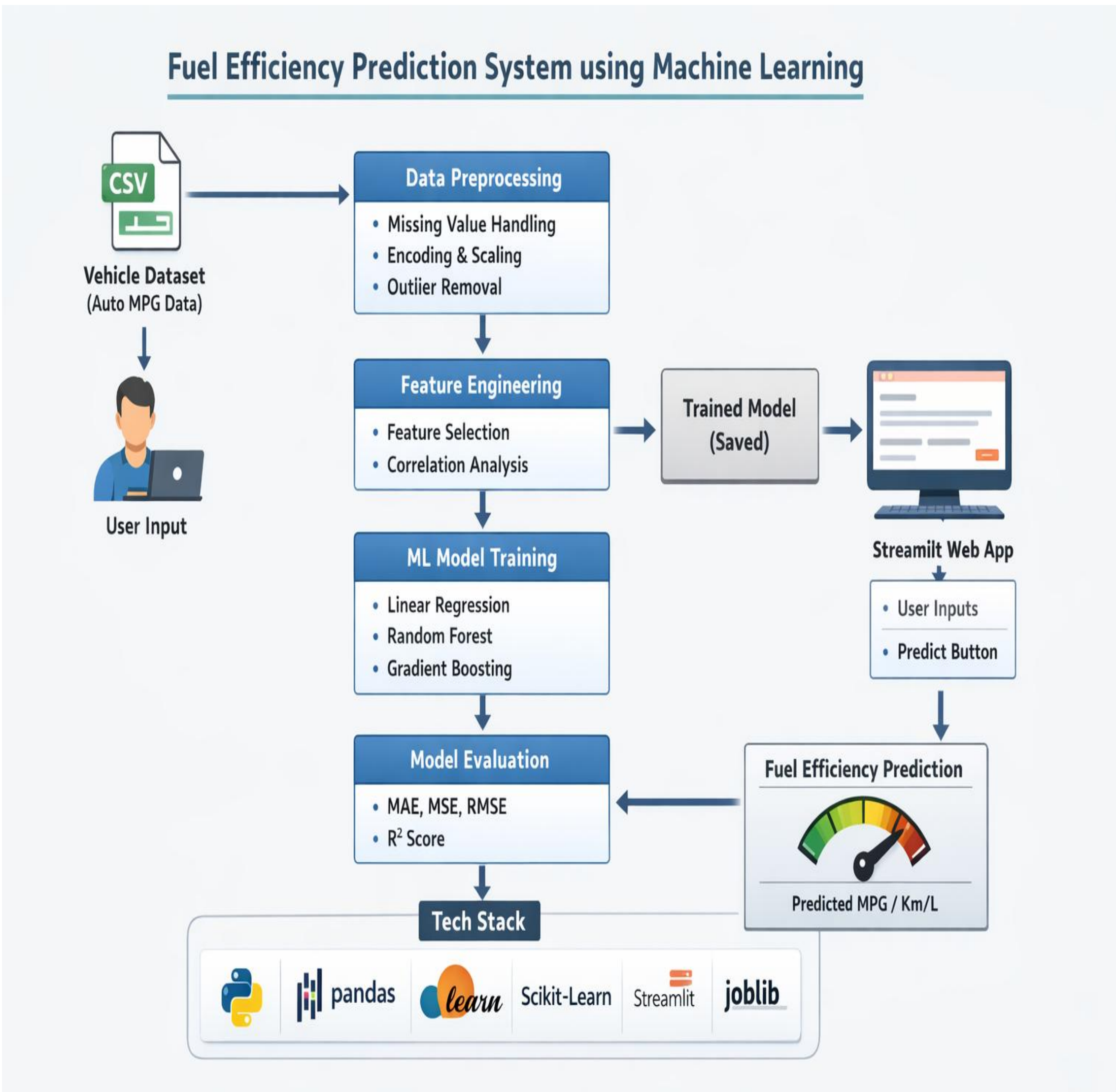

Fig : Flowchart of project

CHAPTER 4: FACILITIES REQUIRED FOR PROPOSED WORK

1. Software Requirements

1. **Programming Languages**:

Python: For data preprocessing, machine learning model development, training, and prediction.
HTML/CSS: For structuring and styling the user interface (used within Streamlit).

2. **Frameworks and Libraries**:

Streamlit: For developing an interactive and user-friendly web interface.
Scikit-learn: For implementing machine learning algorithms and model evaluation.
Pandas & NumPy: For data manipulation, analysis, and numerical computations.
Matplotlib & Seaborn: For visualization of data trends and prediction analysis.

3. **Development Tools**:

Visual Studio Code: For coding, debugging, and project development.
Jupyter Notebook: For data exploration and model experimentation.
Git & GitHub: For version control and project management.

4. **Model Management:**

Joblib: For saving and loading trained machine learning models efficiently.

2. Hardware Requirements

1. User Devices:

- ➢ Personal Computer or Laptop: For data analysis, model training, and application development.
- ➢ Minimum System Requirements:
  - – Processor: Intel i3 or higher
  - – RAM: 8 GB (recommended)

2. Input Devices:

- ➢ Keyboard and Mouse: For development and user interaction.

3. Display Devices:

- ➢ Monitor or Laptop Screen: For visualization and analysis of results.

3. Other Requirements

1. Internet Connectivity:

Reliable internet connection for dataset acquisition, library installation, and cloud deployment.

2. Testing Environment:

Various test inputs and datasets to validate model accuracy and system performance.

3. Documentation Tools:

MS Word / Google Docs for report preparation and project documentation.

CHAPTER 5: REFERENCES

1. S Yoo, J Shin, SH Choi - Scientific Reports, 2025 - nature.com. " Machine learning vehicle fuel efficiency prediction." *Scientific Reports* volume 15,
   Article number: 14815 (2025)
2. Zhang, Y., et al. (2025). *Machine learning-based prediction of vehicle fuel consumption and Efficiency using real-world driving data.* Scientific Reports, Nature.
3. Li, X., et al. (2025). *Data-driven approaches for vehicle fuel efficiency analysis and sustainability assessment.* Sustainability, MDPI, 17(6), 2395.
4. Breiman, L. (2001). *Random Forests.* Machine Learning, 45(1), 5–32.
5. Friedman, J. H. (2001). *Greedy function approximation: A gradient boosting machine.* Annals of Statistics, 29(5), 1189–1232.
6. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction.* Springer.
7. UCI Machine Learning Repository (2023). *Auto MPG Dataset.*
   Retrieved from: https://archive.ics.uci.edu
8. Pedregosa, F., et al. (2011). *Scikit-learn: Machine learning in Python.* Journal of Machine Learning Research, 12, 2825–2830.
9. Streamlit Documentation (2024). *Building Data Apps in Python.*
   Retrieved from: https://docs.streamlit.io